Introduction to Robotics: Kinematics Part 3 – Chassis

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Credit to the book Introduction to Autonomous Mobile Robots, by Roland Siegwart and Illah Nourbakhsh

Differential Pilot: Deprecated

- https://lejos.sourceforge.io/ev3/docs/

Class DifferentialPilot

java.lang.Object
    lejos.robotics.navigation.DifferentialPilot

Deprecated.
use `MovePilot` instead.

```
@Deprecated
public class DifferentialPilot
```

- In Eclipse:

```
public class MeasureMoves {

    public static void main(String [] args) {
        double diam = DifferentialPilot.WHEEL_SIZE_EV3;
        double trackwidth = 10.0; // or 14.5
        DifferentialPilot rov3r = new DifferentialPilot(diam, trackwidth, Motor.A, Motor.B);
```

Solution

- Use Chassis and MovePilot
  - Chassis
    - https://lejos.sourceforge.io/ev3/docs/lejos/robotics/chassis/Chassis.html
  - MovePilot
    - https://lejos.sourceforge.io/ev3/docs/lejos/robotics/navigation/MovePilot.html

Correction to MeasureMoves.java

```
public class MeasureMoves {
   public static void main(String [] args) {
      Wheel wheel1 = WheeledChassis.modelWheel(Motor.A, 56)
                                   .offset(-50);
      Wheel wheel2 = WheeledChassis.modelWheel(Motor.B, 56)
                                   .offset(50);
      Chassis chassis = new WheeledChassis(
               new Wheel[] { wheel1, wheel2 },
                     WheeledChassis.TYPE_DIFFERENTIAL);
      MovePilot rov3r = new MovePilot(chassis);
      rov3r.setLinearSpeed(40);      // unit: mm / sec
      rov3r.setAngularSpeed(45);     // unit: degree / sec

      rov3r.travel(300);      // observe the error
      LCD.drawString("Press ENTER", 0, 3);
      Button.ENTER.waitForPressAndRelease();

      rov3r.rotate(1080);     // observe the error
      Motor.A.close();
      Motor.B.close();
   }
}
```

Correction to PerpetualRov3r.java

- See next page

```java
public class PerpetualRov3r {
public static void main(String[] args) {
  Wheel wheel1 = WheeledChassis.modelWheel(Motor.A, 56).offset(-50);
  Wheel wheel2 = WheeledChassis.modelWheel(Motor.B, 56).offset(50);
  Chassis chassis = new WheeledChassis(new Wheel[] { wheel1, wheel2 },
          WheeledChassis.TYPE_DIFFERENTIAL);
  MovePilot rov3r = new MovePilot(chassis);
  NXTUltrasonicSensor us = new NXTUltrasonicSensor(SensorPort.S1);
  SensorMode distMode = us.getMode("Distance");
  double distance = 1.0;

  rov3r.setLinearSpeed(40);      // unit: mm / sec
  rov3r.setAngularSpeed(45);     // unit: degree / sec
  rov3r.forward();
  while (!Button.ESCAPE.isDown()) {// hold the button for a while
    float[] sample = new float[distMode.sampleSize()];
    distMode.fetchSample(sample, 0);
    distance = sample[0];

    LCD.drawString("DIST:      ", 0, 2);
    LCD.drawString(Double.toString(distance), 5, 2);
    Delay.msDelay(400);

    if (distance < 0.30 && distance > 0.1) {          // draw back and turn
      rov3r.travel(-200);        // unit: mm
      rov3r.rotate(45);
      rov3r.forward();
    }
    Delay.msDelay(100);
  }
  us.close();
  Motor.A.close();
  Motor.B.close();
  }
}
```

Self Test

- Correct the deprecated parts in Ackerbot.java